

ALEX CHEN

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EDUCATION

Bachelor of Technology in Computer Science & Engineering | Chitkara University

CGPA: 9.2 / 10.0 • Expected June 2027 • Core: Distributed Systems, Database Management, Algorithms, Networks

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Go, SQL (PostgreSQL, MySQL)

Backend & Architecture: Node.js, Express, REST APIs, GraphQL, Redis, Distributed Systems, Microservices

Frontend & UI: React.js, Next.js (App Router), Tailwind CSS, Framer Motion, HTML5, WebSockets

Cloud & DevOps: Microsoft Azure, Docker, Kubernetes, Git, CI/CD (GitHub Actions), Vitest, Jest

ENGINEERING PROJECTS

Distributed Order & Inventory Processing Platform | Node.js, Express, PostgreSQL, Redis, Docker

- **Database Schema & Integrity:** Architected relational transaction schema in PostgreSQL with strict ACID guarantees, utilizing row-level locking (SELECT ... FOR UPDATE) to prevent double-booking during flash-sale concurrency spikes.
- **Caching & Invalidation:** Integrated Redis read-through caching for high-traffic inventory endpoints, slashing average query response latency by 68% (from 145ms to 46ms) with atomic write-through invalidation.
- **Fault Tolerance:** Implemented asynchronous order fulfillment pipeline with dead-letter queue retries, processing 3,500+ simulated concurrent orders with zero data loss.

Real-Time High-Throughput Metrics Dashboard | Next.js, React, WebSockets, Apache Kafka, InfluxDB

- **Streaming Telemetry:** Engineered real-time browser visualizer handling 8,000+ incoming metric events/second over duplex WebSockets without DOM frame dropping (60 FPS).
- **State & Aggregation:** Designed windowed client-side buffering and decimation algorithms to render time-series plots without memory leaks across prolonged sessions.

Rehearse — Voice-Driven Autonomous AI Interview Coach | Next.js 14, Azure Speech, Azure OpenAI (GPT-4o)

- **Speech & Reasoning Loop:** Integrated continuous Azure Speech-to-Text streaming and 24kHz Neural TTS with SSML prosody; connected to GPT-4o for contextual multi-turn technical trade-off probing.
- **Responsible AI Architecture:** Architected zero-cloud local WebRTC camera stream for 100% video privacy and serverless ephemeral STS token authentication.

EXPERIENCE

Software Engineering Intern | CloudTech Innovations (May 2025 – August 2025)

- **API Optimization:** Refactored 14 core REST endpoints, implementing query pagination and Redis caching that cut P95 latency from 320ms to 95ms.
- **CI/CD & Testing:** Authored automated unit and integration test suites achieving 88% branch coverage, reducing production hotfix releases by 40%.